

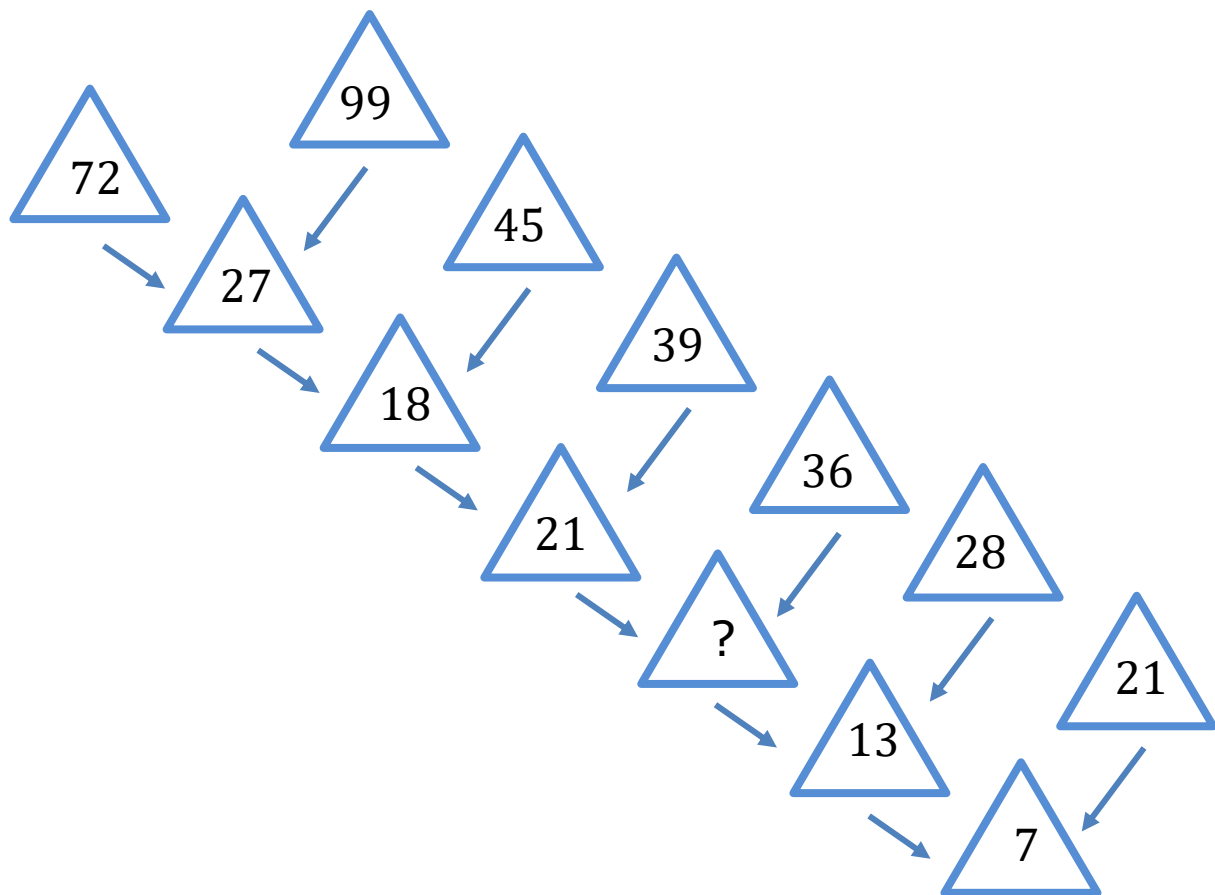
Unit 11, Lesson 5: Generating Equivalent Algebraic Expressions

Benchmark

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

Learning Target

- ☐ I can apply properties of operations to generate equivalent expressions.

11.5.1 Warm-Up: Missing Number

1. What is the first number you tried for the missing number?
2. Did your first number work for the rest of the pattern?
3. What strategy did you try if your first number did not work?

11.5.2 Guided Instruction: Generating Equivalent Algebraic Expressions

1. Consider the following expressions.

$8 \cdot x + 8 \cdot 9$

$8(x + 9)$

$8x + 9$

- a. Which of these expressions are equivalent?
- b. Which property of operation justifies that the expressions are equivalent?

2. Write two equivalent expressions. Name the property or properties you used to create each equivalent expression.
3. For each expression, use the associative and commutative properties to generate an equivalent expression.

Original Expression	Associative Property	Commutative Property
$(3x + 5) + 4z$		
$8 \cdot (-1 \cdot 13d)$		

4. Rewrite $-11b + 3(10k + 1)$ using the distributive property.
5. Rewrite $-11b + 3(10k + 1)$ using the commutative property of multiplication.
6. Rewrite $-11b + 3(10k + 1)$ using the commutative property of addition.
7. Makayla and Soledad both rewrote $(-5h + 9) \cdot -14$ using the commutative property of multiplication. Verify that their expressions are equivalent by substituting a value in for h . Show your work in the table.

Makayla	Soledad
$-14(-5h + 9)$	$(h \cdot -5 + 9) \cdot -14$
$h = \rule{1cm}{0.4pt}$	$h = \rule{1cm}{0.4pt}$

There are some fundamental definitions that are necessary to know for working with algebraic expressions.

A **term** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The expression $5 - 3y$ has two terms.

5		$-3y$
term 1		term 2

The term, 5, is a constant term because it is a fixed numerical value with no variable.

The term, $-3y$, is a variable term with a coefficient being multiplied by the variable.



A **coefficient** is the number or constant that multiplies a variable in an algebraic expression. If no number is specified, the coefficient is 1.

Like terms are terms with the exact same variable(s) raised to the same exponent. Constant terms are like terms.

Like terms can be combined when adding and subtracting.

8. Rewrite each expression by combining like terms and applying any properties of operations.

a. $5x - 3x =$ _____

b. $-2y + 4 + 4y - 1 =$ _____

c. $6x + 2(y - x) =$ _____

11.5.3 Try It: Writing Equivalent Algebraic Expressions

1. Select the expressions that are equivalent to $(2x + 7) + (5x - 3)$.

- ☐ $(5x - 3) + (7 + 2x)$
- ☐ $(3 - 5x) + (7 + 2x)$
- ☐ $7 + (5x + 2x) - 3$
- ☐ $(7 + 5x) + (2x - 3)$
- ☐ $5x + (7 + 2x) - 3$
- ☐ $7x + 10$
- ☐ $7x + 4$
- ☐ $11x$

2. Write two equivalent algebraic expressions for each expression below.

a. $(4 + 9) \cdot x$

b. $3t \cdot 4$

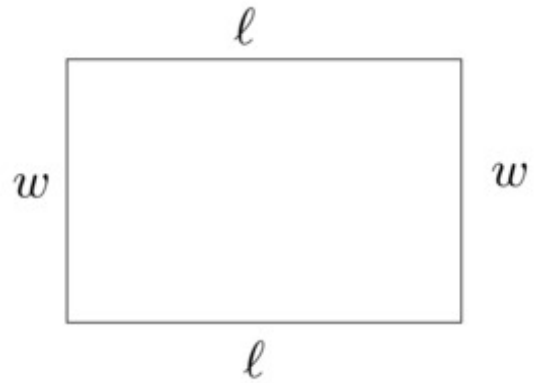
c. $-7x + 3 - 14$

d. $3(-9p - 4) + 7p$

11.5.4 Your Turn!

1. The students in Ms. Justino's class are writing expressions for the perimeter of a rectangle of side length, ℓ , and width, w . The expressions for four students are in the table.

Student	Expression
De'Andre	$2(\ell + w)$
Summer	$\ell + w + \ell + w$
Abm	$2\ell + w$
Luzarie	$2\ell + 2w$



Ms. Justino told the group that one of the expressions is not equivalent. Write a short explanation to the student to show why their expression is not equivalent to the other expressions.

Mmeans
I
Sstart
To
Acquire
Knowledge
Experience
Skills

2. Match equivalent expressions by drawing an arrow. If an expression does not have a match, write an equivalent expression to match it in one of the blank boxes provided.

Column A
$-8(x + 3)$
$-8x \cdot 3$
$(8x - 3x) \cdot 2$
$(3 + -8x) + 2$
$2y + -8x$

Column B
$2 + (-8x + 3)$
$3 \cdot -8x$
$-24 - 8x$

11.5.5 Wrap-Up: Error Analysis

Tangela was asked to circle all of the expressions equivalent to $6(y + 1)$.

Her work is shown below.

$6y + 1$	$6y + 7$	$6(y) + 1(y)$	$6(y) + 6(1)$	$6y + 6$
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1. Determine if Tangela correctly identified all of the equivalent expressions. If she is missing an expression, circle the expression(s) she is missing.
2. If Tangela made an error, then explain her error and list the equivalent expressions.

